

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Unit Test 2

Unit Test 2

Class: Class 9

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: Data Structures

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

a. What will be the output?

```
d = {"a": [1, 2], "b": [3, 4]}
d["a"].append(5)
print(d["a"])
```

- a. [1, 2]
- b. [1, 2, 5]
- c. [5, 1, 2]
- d. Error

b. Which method is used to sort a list in place?

- c. sorted()
- d. sort()
- e. order()
- f. arrange()

c. What is the output?

```
t = ([1, 2], [3, 4])
t[0].append(5)
print(t)
```

- a. ([1, 2], [3, 4])
- b. ([1, 2, 5], [3, 4])
- c. Error (tuples are immutable)
- d. ([5, 1, 2], [3, 4])

d. Which of the following creates a shallow copy of a list?

- e. copy = list
- f. copy = list[:]
- g. copy = list.copy()
- h. Both B and C

e. What does set([1, 2, 2, 3, 3, 3]).pop() return?

- f. An arbitrary element from the set

- g. The last element added
- h. Always 1
- i. Error

Section II: Short Answer (5 x 1 = 5 marks)

- a. Write the output of:

```
nums = [3, 1, 4, 1, 5, 9, 2, 6]
print(sorted(nums, reverse=True)[:3])
```

- b. Write a program to merge two dictionaries. If a key exists in both, sum the values.

- c. What will be the output and why?

```
lst = [1, 2, 3]
lst2 = lst
lst2.append(4)
print(lst)
```

- d. Differentiate between `pop()` and `remove()` for dictionaries with examples.

- e. Write a program to flatten a nested list `[[1, 2], [3, 4], [5, 6, 7]]` into a single list.